

Implicit Neural Representations for Efficient and Flexible Whole Slide Image Compression

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Image Processing Final Project

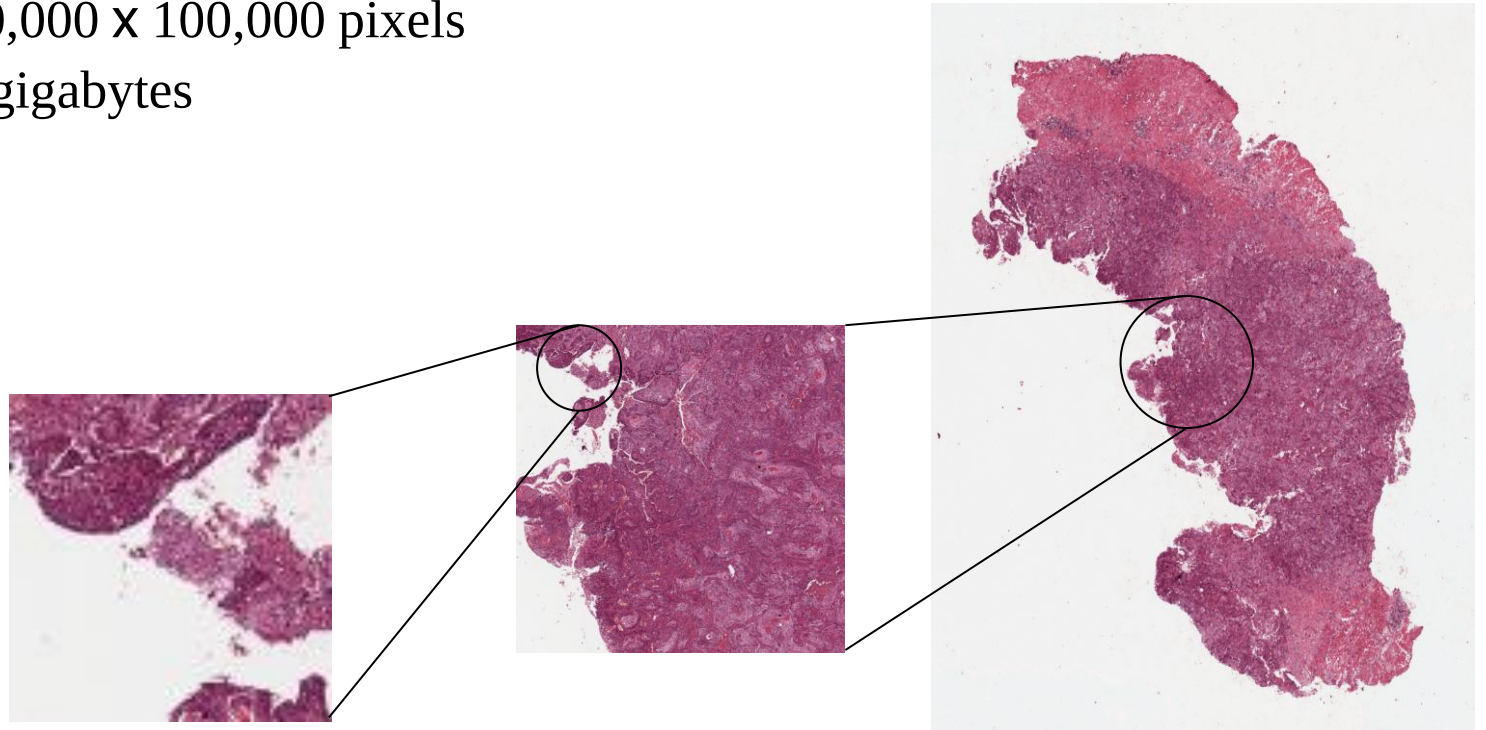
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Outline

- **Background, Motivation and Overview**
- **Methods**
- **Experiments and Results**
- **Conclusion**

Background and Motivations

- **Whole Slide Imaging (WSI) enables high-resolution digital pathology:**
 - Typical WSIs exceed 100,000 x 100,000 pixels
 - Generate files of several gigabytes
 - Serious challenges for:
 - ☐ Storage
 - ☐ Transmission
 - ☐ Interactive analysis.



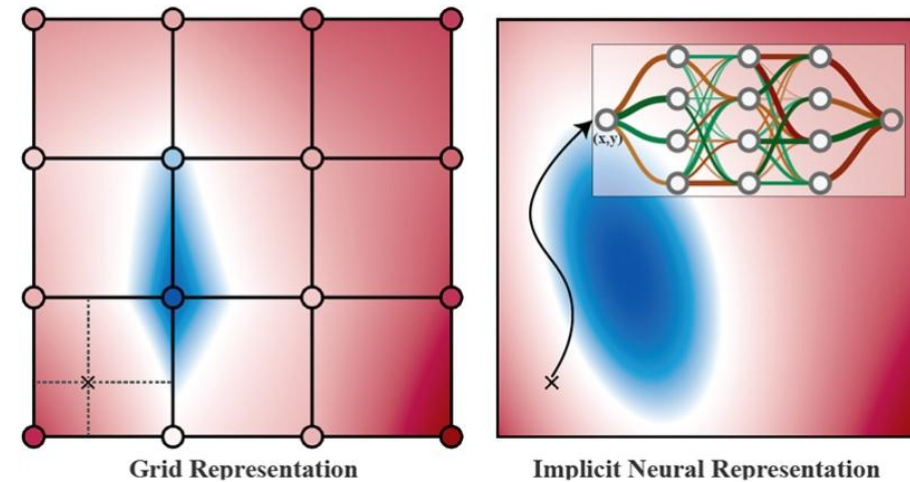
Background and Motivations

- **Traditional Methods:**

- JPEG2000



- Implicit Neural Representations (INRs)



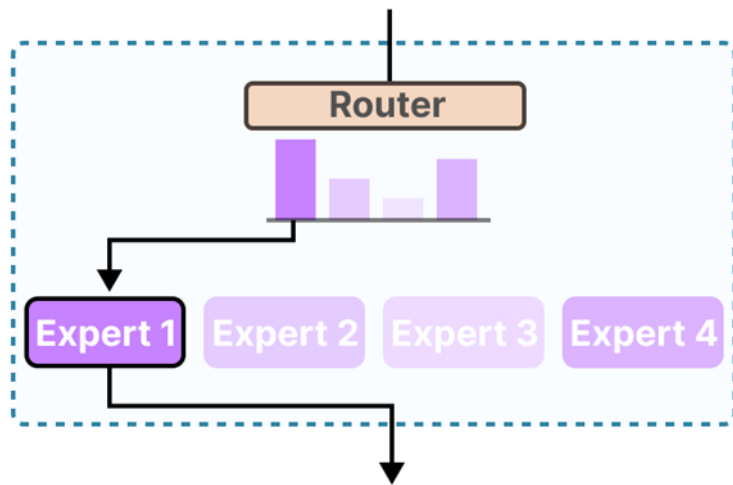
Background and Motivations

- **WSI data poses unique challenges:**
 - **Extreme resolution**
 - **Large spatial variation**
 - **The coexistence of densely structured tissue**
 - **Large blank areas.**

Overview

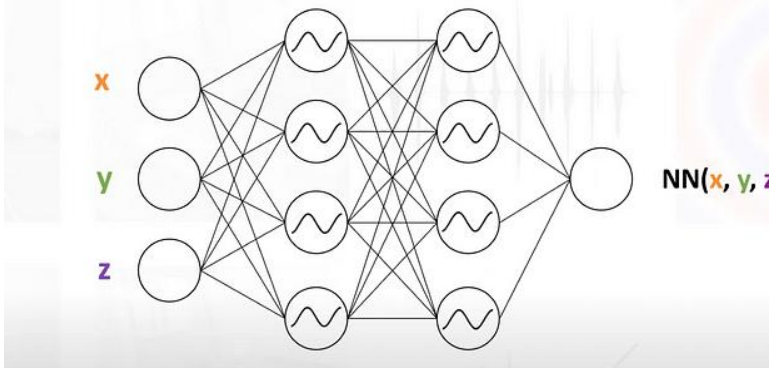
- **Our Proposed Methods:**

- Mixture-of-Experts (MoE) INR

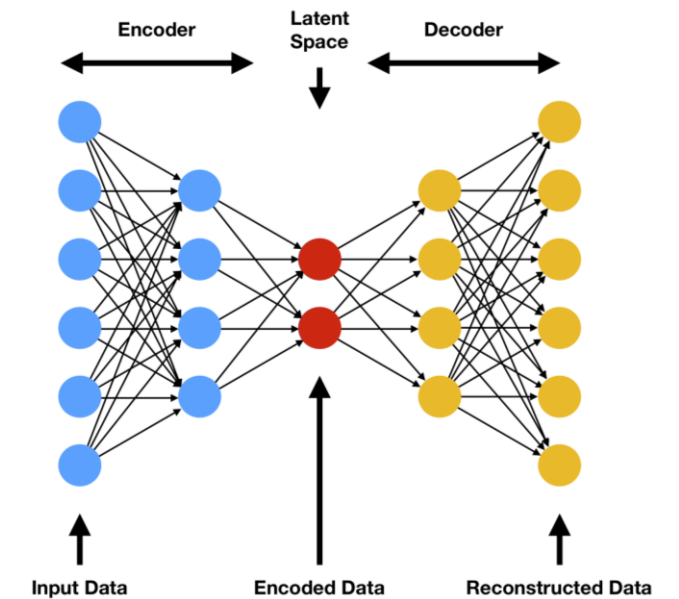


- Frequency based Method

SIREN: Sinusoidal Representation Networks

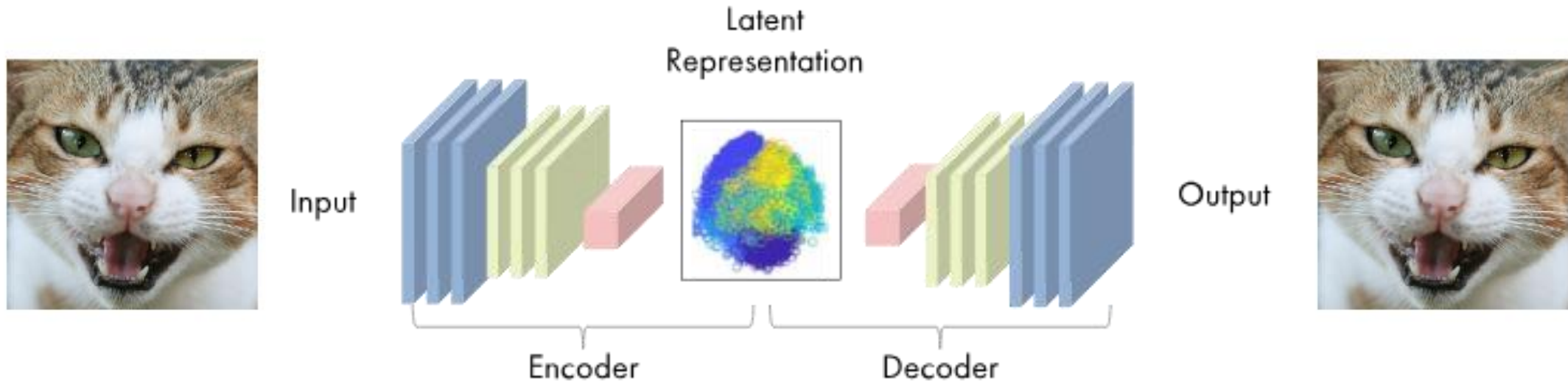


- AutoEncoder Method



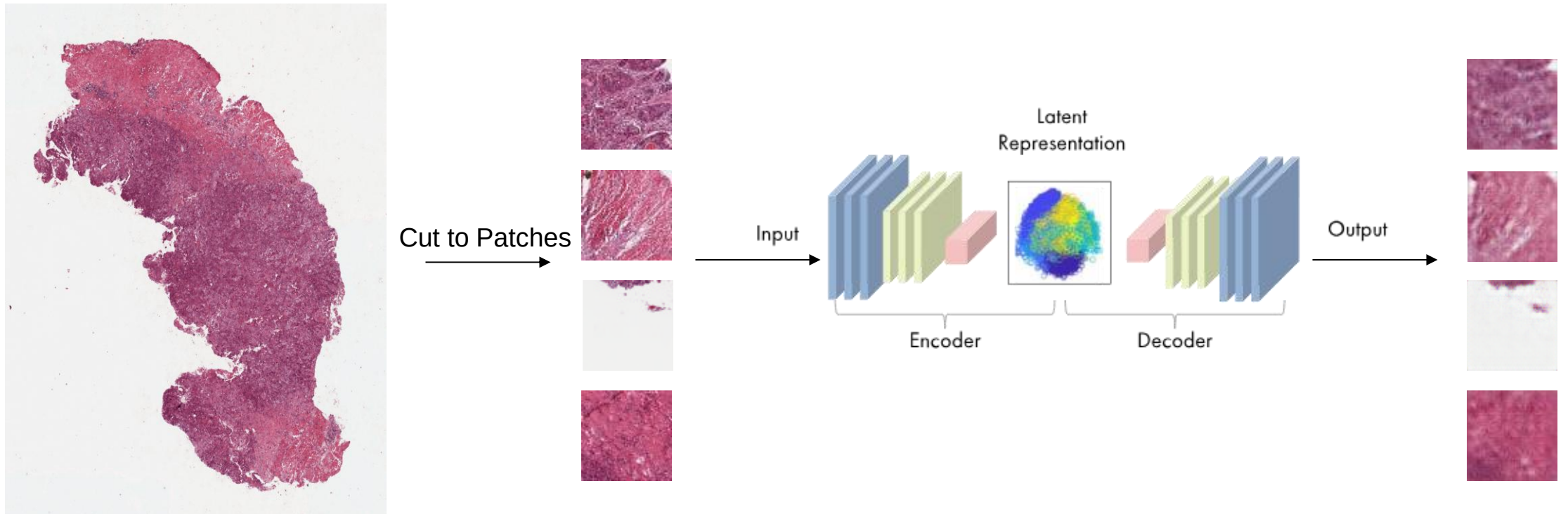
Methods

- AutoEncoder Based Method:
 - Model Architecture



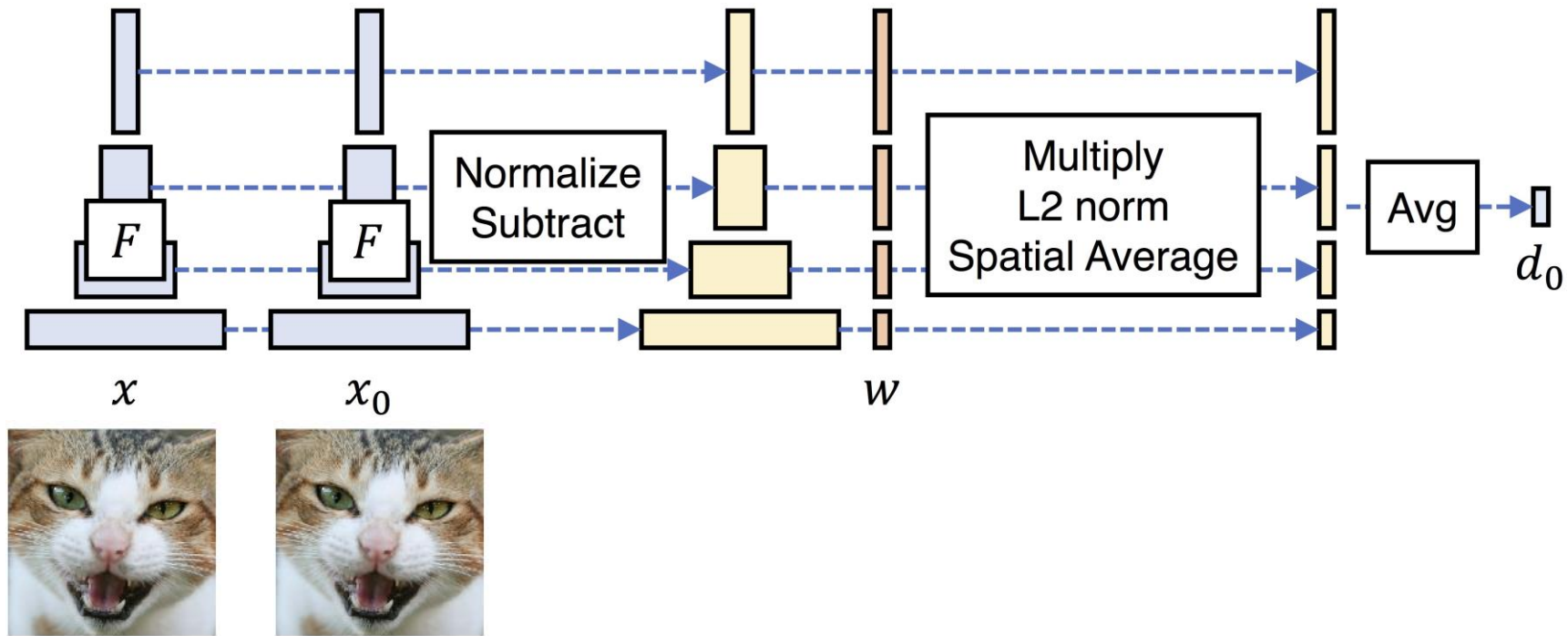
Methods

- AutoEncoder Based Method:
 - Patch-wise Implementation:



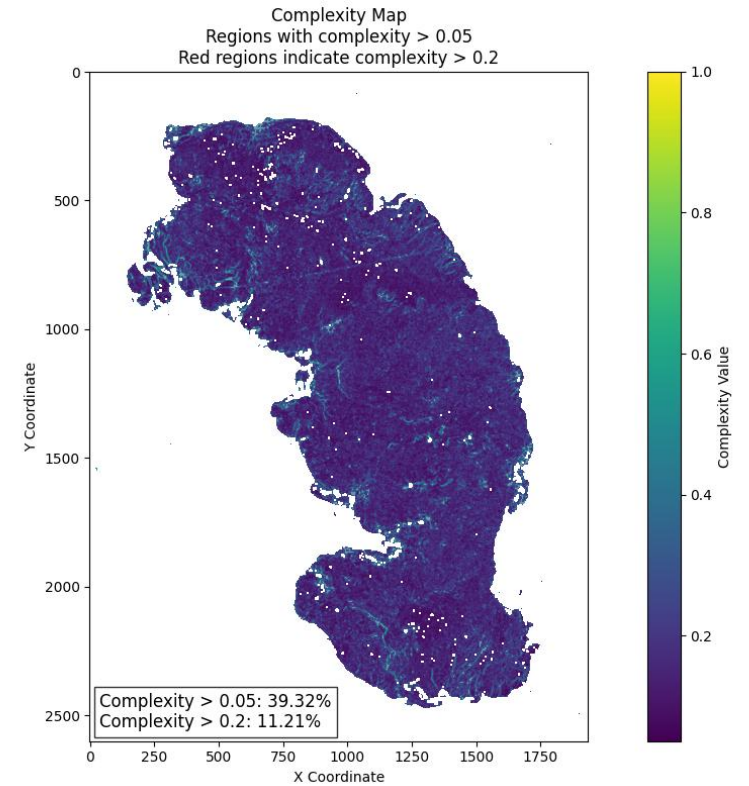
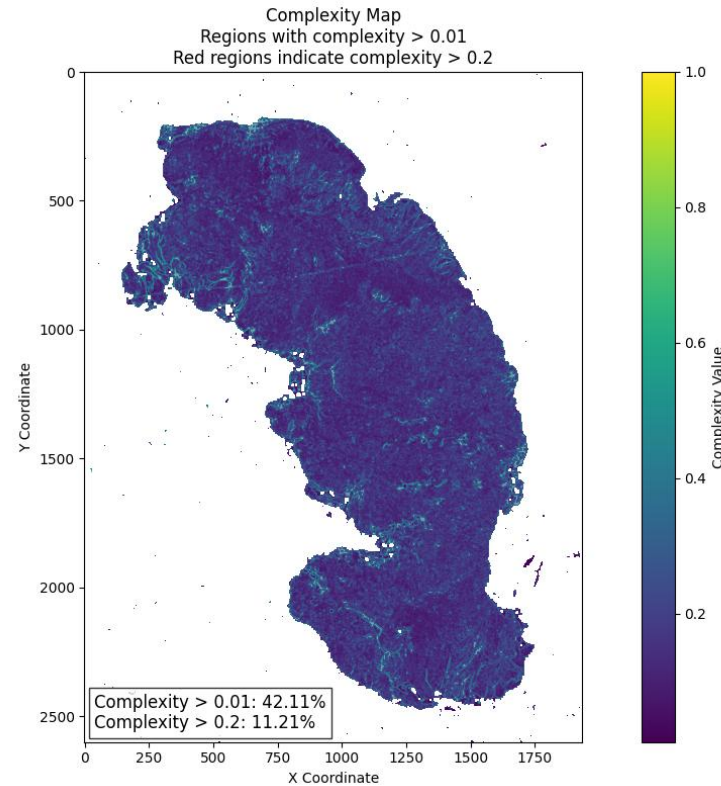
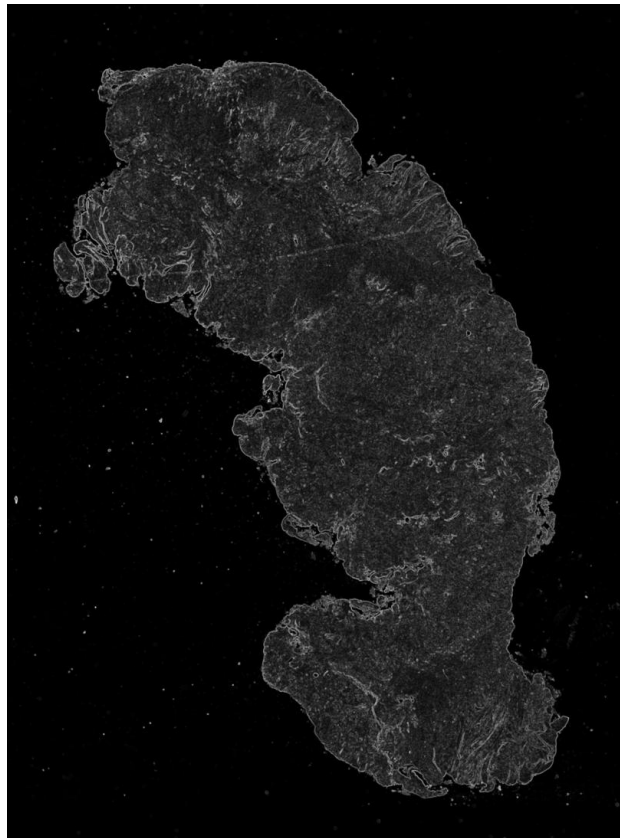
Methods

- AutoEncoder Based Method:
 - LPIPS Loss + L2 Loss



Methods

- Mixture-of-Experts (MoE) :



Methods

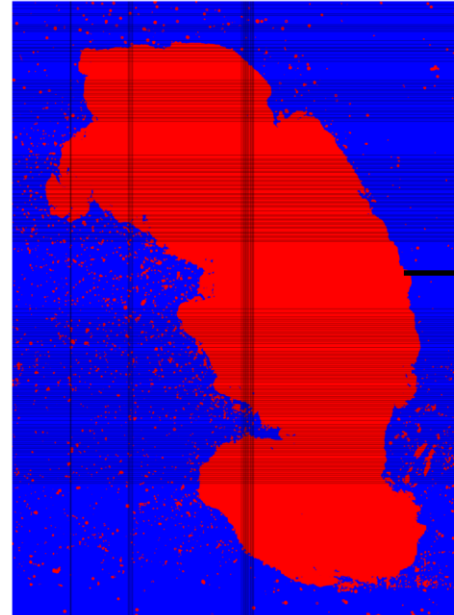
- Mixture-of-Experts (MoE) :

- Adaptive Sampling

$$P(\mathbf{x}) = \frac{c(\mathbf{x})}{\sum_{\mathbf{x}'} c(\mathbf{x}')}$$

- Multi-Branch Encoding

$$\mathbf{z} = \left[\phi_{\text{pos}}(\mathbf{x}), \phi_{\text{reg}}(\mathbf{x}), \sum_{l=1}^3 \alpha_l(\mathbf{x}) \cdot \phi_{\text{scale}}^{(l)}(\mathbf{x}) \right]$$



Highly textured edge locations are carried into more training batches

Methods

- Mixture-of-Experts (MoE) :

- Multi-Branch Encoding

$$\mathbf{z} = \left[\phi_{\text{pos}}(\mathbf{x}), \phi_{\text{reg}}(\mathbf{x}), \sum_{l=1}^3 \alpha_l(\mathbf{x}) \cdot \phi_{\text{scale}}^{(l)}(\mathbf{x}) \right]$$

- **Positional Encoding:** Uses classic sine/cosine series expansion, covering a wide frequency range from low to high
- **Area Encoding:** A small MLP directly processes the 2D coordinate and outputs semantic features of the region. “what kind of local structure it might represent.”
- **Scale Encoding:** Corresponding to three pyramid resolutions, there are three separate MLP branches, each generating a feature vector and a scalar relevance score.

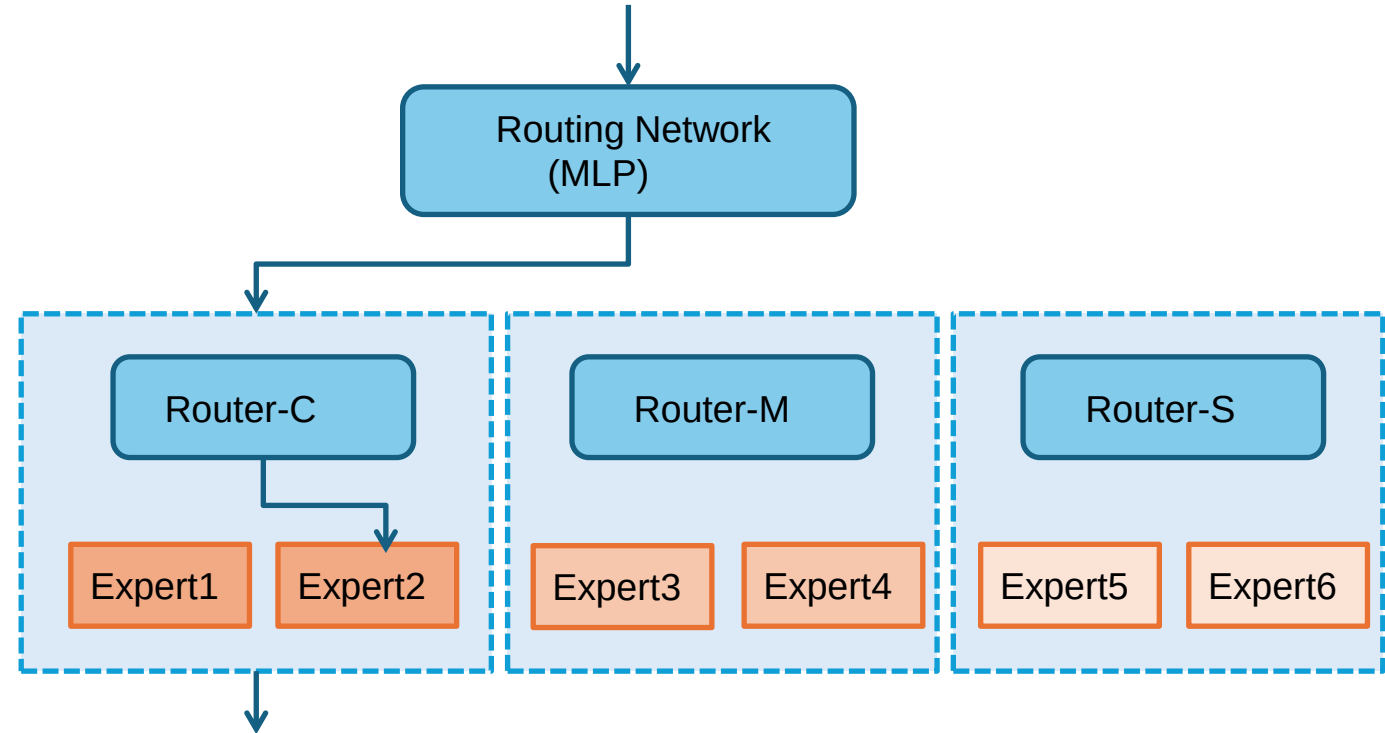
Methods

- Mixture-of-Experts (MoE) :

- Expert Routing

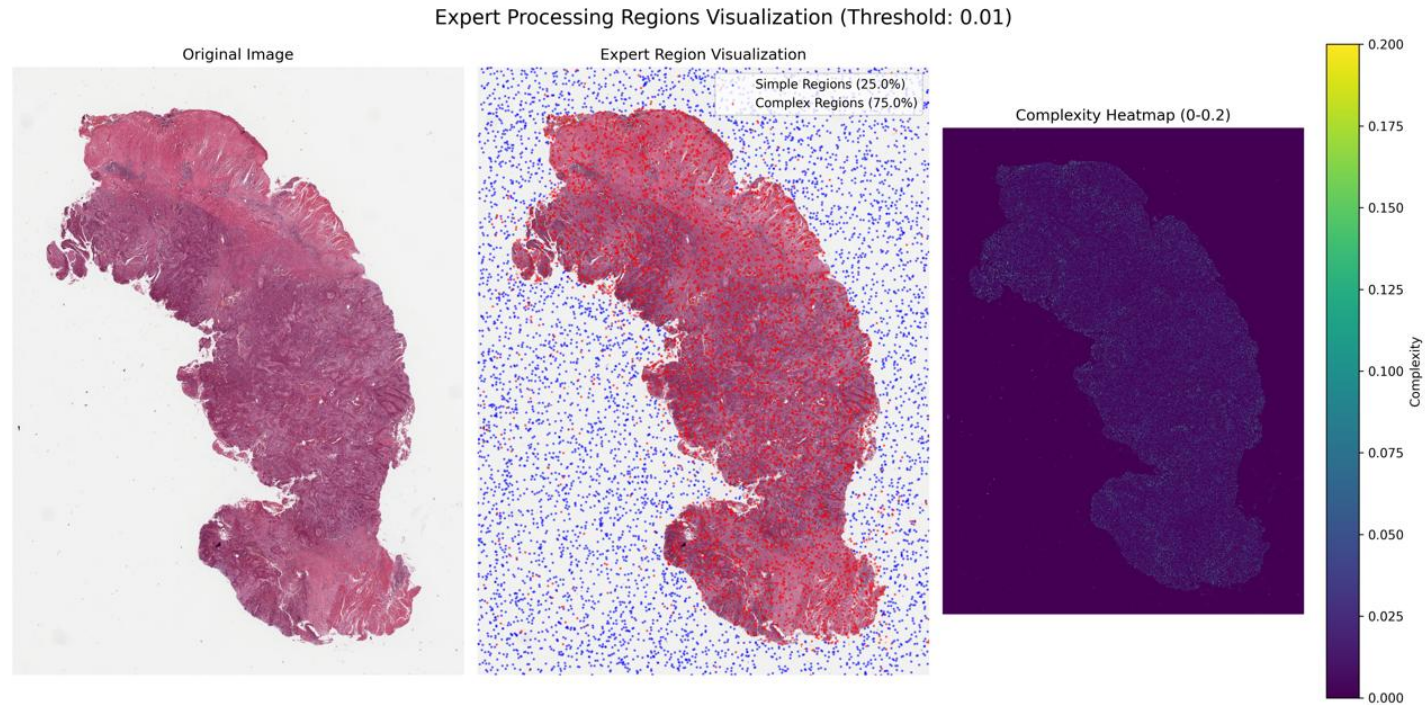
$$g'_i = \begin{cases} 2g_i & \text{if } i \in \mathcal{S}, c(\mathbf{x}) \leq 0.01 \\ g_i & \text{if } i \in \mathcal{M}, 0.01 < c(\mathbf{x}) \leq 0.05 \\ 2g_i & \text{if } i \in \mathcal{C}, c(\mathbf{x}) > 0.05 \end{cases}$$

$$\hat{I}(\mathbf{x}) = \sum_{i=1}^6 w_i \cdot f_i(\mathbf{x})$$



Methods

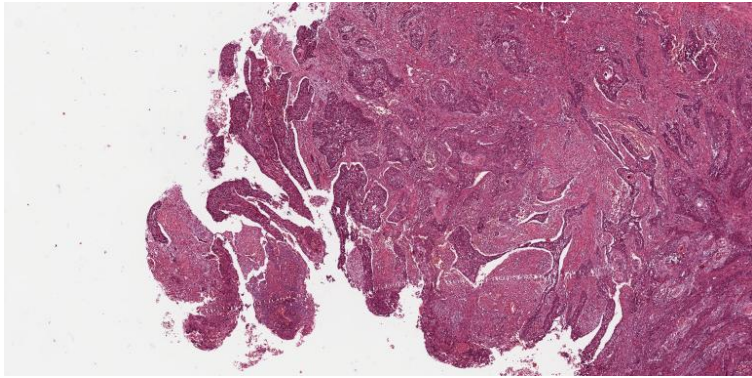
- Mixture-of-Experts (MoE) :



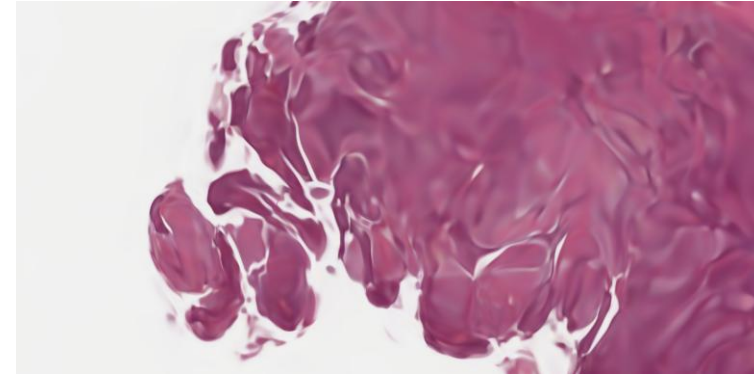
$$\mathcal{L} = \text{MSE} + \lambda \cdot \mathcal{H}(\mathbf{w}) = \|\hat{I}(\mathbf{x}) - I_{\text{gt}}(\mathbf{x})\|^2 - \lambda \sum_{i=1}^6 w_i \log w_i,$$

Methods

- Frequency-Based Method :



(a) Original Image

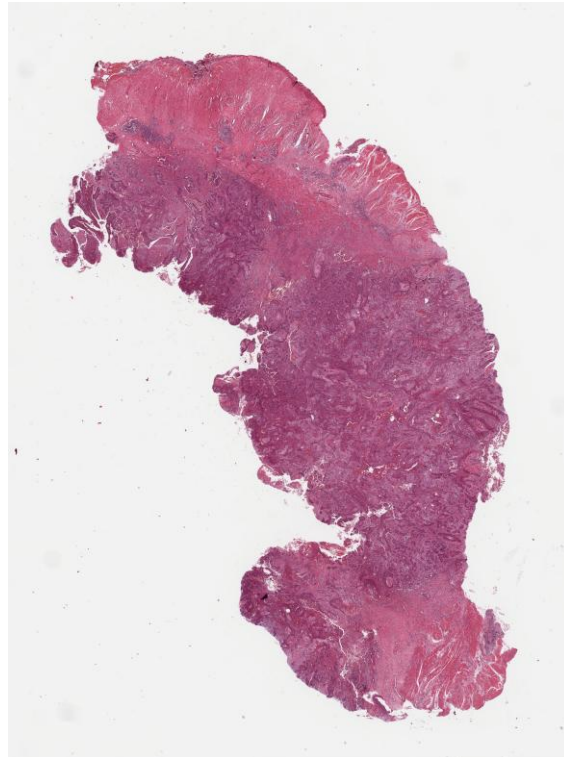


(b) Baseline

The original image contains rich details, i.e., high-frequency components, which are largely missing in the reconstructions of baseline methods.

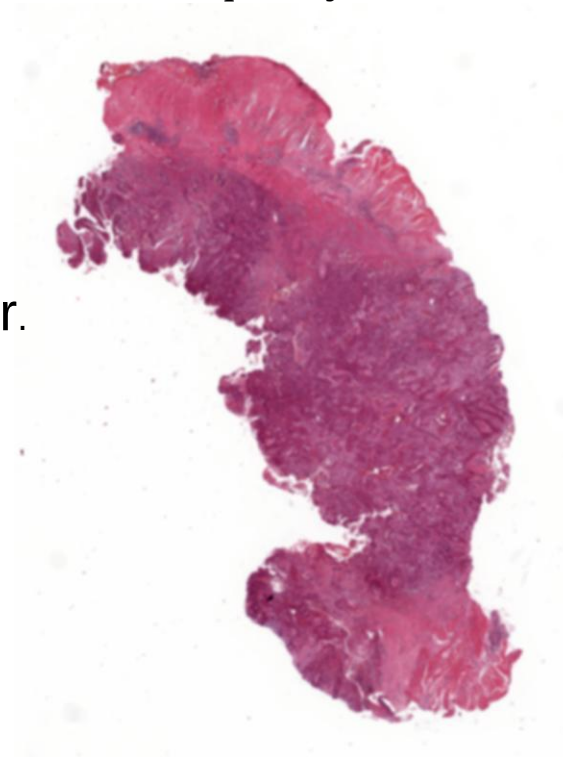
Methods

- Frequency-Based Method :



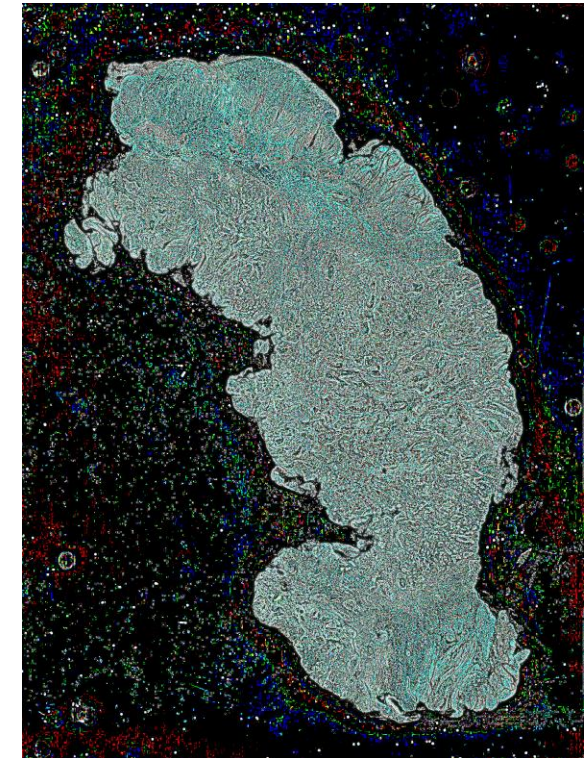
original_image.png

Gaussian filter.
→



low-frequency component

$$\text{Low-frequency} = G \sigma * I$$

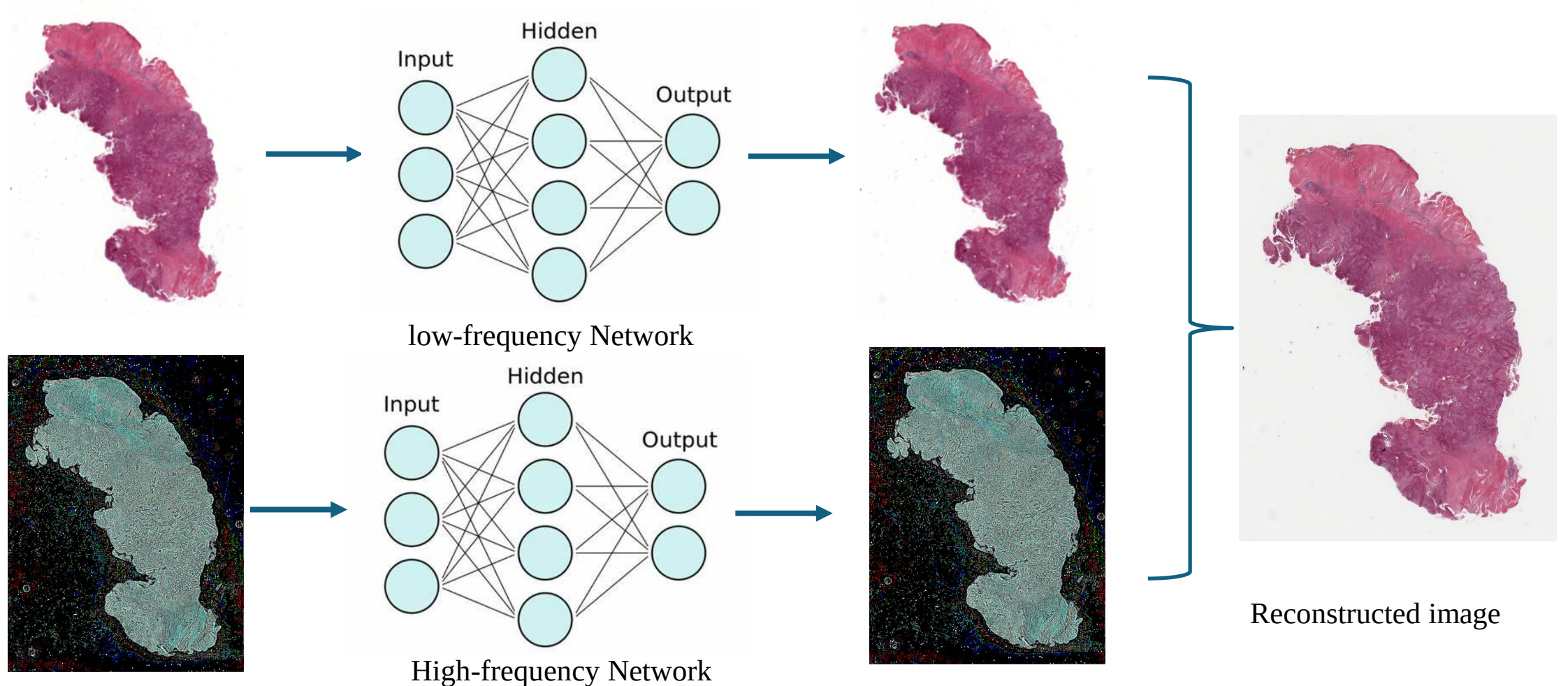


High-frequency component

$$\text{High-frequency} = I - \text{Low-frequency}$$

Methods

- Frequency-Based Method :



Methods

Low-Frequency Network (Shallow CoordNet)

- Uses fewer residual blocks (num_res=1)
- Captures smooth structures and coarse intensity variations
- Emphasizes global consistency over detail

High-Frequency Network (Deep CoordNet)

- Uses more residual blocks (e.g., num_res=10)
- Learns fine textures and sharp edges
- Focuses on modeling detailed high-frequency components

Thank You!

Q&A